Fundamental Parallelepiped and Tiling

Lattice generated by basis: $B = [\mathbf{b}_1\ \mathbf{b}_2]$

$\mathbf{b}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix},\ \mathbf{b}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\mathcal{P}(B) = \{\alpha_1 \mathbf{b}_1 + \alpha_2 \mathbf{b}_2 : 0 \le \alpha_1, \alpha_2 < 1\}$

Notice: Boundaries defined by half-open intervals.

Tiling: $\mathbb{R}^2 = \bigcup_{\mathbf{y} \in \Lambda} (\mathbf{y} + \mathcal{P}(B))$

Example of unique decomposition shown:

$\mathbf{x} = \mathbf{y} + \mathbf{p}$ where $\mathbf{y} \in \Lambda,\ \mathbf{p} \in \mathcal{P}(B)$.

